# Reproducing the Visual Search Advantage for Illusory Faces: A Behavioral and Diffusion Decision Modeling Study

Xiemohan Yue and Renyi Yang

Grenoble INP - Phelma

VPMDLP11 Lab project I

Marion Dohen, Nathalie Guyader, and Ronald Phlypo

May 20, 2026

## Abstract

This report presents a Python/Pygame reproduction and computational extension of the visual search experiments reported by Keys et al. (2021), which investigated whether illusory faces in everyday objects confer a visual search advantage. Behavioral results replicated the primary findings of the original study: reaction times increased with set size, target-absent trials were slower than target-present trials, and illusory faces were detected more efficiently than non-face objects. In Experiment 2, real faces produced the fastest responses overall.

To investigate the underlying decision mechanisms, diffusion decision modeling (DDM) was applied to reaction time distributions across conditions. Model comparisons indicated that target-present trials were primarily explained by changes in drift rate, suggesting reduced evidence accumulation efficiency at larger set sizes. Real faces showed the highest drift rates, illusory faces showed intermediate drift rates, and non-face objects showed the lowest drift rates. By contrast, target-absent trials were better explained by increases in boundary separation, consistent with increasingly cautious search termination decisions.

These findings demonstrate that illusory faces confer a measurable perceptual advantage during visual search and show how DDM can dissociate evidence accumulation from response caution in face-related visual search tasks.


*Keywords:* visual search, illusory faces, pareidolia, diffusion decision model, reaction time

# Reproducing the Visual Search Advantage for Illusory Faces: A Behavioral and Diffusion Decision Modeling Study

## Introduction

### Background: Face Detection and Visual Search

The human visual system prioritizes the detection of faces in cluttered environments. This attentional bias reflects both the ecological importance of face detection in human social cognition and the evolutionary cost of missing a conspecific in a naturalistic setting. Substantial evidence demonstrates that human faces are detected faster and more efficiently than non-face objects in visual search tasks (e.g., Hershler and Hochstein (2005) and Wolfe (2001)).

A fascinating and well-documented phenomenon is *face pareidolia*—the spontaneous perception of faces in inanimate objects, such as electrical sockets, toast, or tree trunks. Critically, these illusory face perceptions occur despite the absence of canonical facial features (e.g., skin texture, stereotypic face geometry). This raises an intriguing question: does the illusory face percept leverage the same rapid detection mechanism as real faces?

### The Keys et al. (2021) Study

Keys et al. (2021) addressed this question using a classic visual search paradigm implemented in MATLAB with Psychtoolbox-3 (PTB-3). They conducted two experiments:

1. **Experiment 1 (Grid Search)**: Participants searched for objects containing illusory faces among homogeneous distractors (all items from the same object category). Set sizes ranged from 16 to 64 items arranged on an invisible $8 \times 8$ grid.

2. **Experiment 2 (Circular Search)**: Participants searched for three target types (real human faces, illusory faces, and non-face control objects) among heterogeneous distractors arranged in circular arrays of 4, 8, or 16 items.

Both experiments found a robust search advantage for illusory faces, suggesting that the rapid face-detection mechanism is tuned to high-level facial *configuration* rather than to the low-level features that distinguish real faces from other objects.

## This Reproduction

The present work translates the MATLAB/PTB-3 implementation into Python with Pygame, a modern, cross-platform graphics library. This choice enables:

- **Platform independence**: The code runs identically on Windows, macOS, and Linux.
- **Open-source accessibility**: No proprietary software is required.
- **Reproducibility**: The implementation is public and version-controlled.

The goal is *strict methodological fidelity*: we replicate all timing parameters, spatial layouts, stimulus presentation protocols, randomization schemes, and data recording formats of the original study.

To clarify the mechanism behind any search advantage, we supplemented descriptive performance measures with diffusion decision modeling (DDM; Ratcliff and McKoon (2008)). Standard analyses of mean reaction times and accuracy cannot disentangle whether faster responses reflect more efficient evidence accumulation or shifts in response caution. DDM addresses this limitation by decomposing choice and reaction time distributions into psychologically interpretable components, allowing us to test whether set size and target type influence evidence quality, decision thresholds, or nondecision processes. This modeling approach is particularly appropriate for visual search, where increasing array size may reduce evidence efficiency, increase cautiousness, or both.

## Present Hypotheses

Based on the original findings of Keys et al. (2021), we expected illusory-face targets to produce faster and more accurate responses than matched nonface objects during visual search. In Experiment 2, we further expected real faces to produce the greatest search advantage overall.

At the computational level, we hypothesized that these behavioral advantages would primarily be reflected in drift rate differences within the diffusion decision model, indicating more efficient evidence accumulation for face-related stimuli. Specifically, we expected real faces to produce the highest drift rates, illusory faces intermediate drift rates, and nonface objects the lowest drift rates.

For target-absent trials, we expected increasing set size to promote more conservative search termination processes, reflected by increases in boundary separation parameters.

# Method

## Experiment 1: Grid Layout (Homogeneous Distractors)

### *Stimulus Presentation*

Each trial in Experiment 1 involves the following sequence:

1. **Target Image**: A single target image (either pFace or nonFace) is displayed in the center of the screen for 1600 ms.
2. **Fixation**: A fixation cross is displayed for a randomized duration between 400 and 600 ms (uniform random distribution).
3. **Search Array**: A $8 \times 8$ grid of images is displayed. Participants respond via keyboard (left/right arrow keys or letter keys) indicating whether the target is present or absent. The array remains visible until response or a timeout of 15,000 ms.
4. **Feedback**: A colored fixation cross (green for correct, red for incorrect or timeout) is displayed for 250 ms.

### *Experimental Factors*

Three factors are manipulated in a fully-crossed design:

| **Factor** | **Levels** | **N** |
|---|---|---|
| Target Type (pFace vs. nonFace) | 2 | |
| Target Presence | 2 | |
| Set Size | 3 (16, 32, 64) | |
| Object Categories | 26 | |
| Trial Types per Category | 12 | |
| **Total Trials** | | 312 |

Twenty-six participants completed Experiment 1. Participants were university students aged 20–22 years. Gender information was not recorded. All participants had normal or corrected-to-normal vision and provided informed consent. The dataset contains 1352 trial rows, with 52 trials per participant.

### *Trial Type Definition*

The 12 trial types per category correspond to the orthogonal combination of target type, target presence, and set size: Types 1–3 present pFace (set sizes 16/32/64), Types 4–6 absent pFace, Types 7–9 present nonFace, Types 10–12 absent nonFace.

### *Stimulus Yocking (Critical Detail)*

A crucial aspect of Experiment 1 is the *yoked stimulus design*: all distractor images in a given trial come from the **same object category** as the target. For example, if the target is Category 5 (cheese graters) with pFace variant, all 15, 31, or 63 distractor images are unique cheese graters from Category 5. The target image occupies one of the 16/32/64 grid positions, randomly selected. The remaining positions are filled by random distractors from the same category. This ensures a *homogeneous visual context*, maximizing the influence of the target's face-like configuration versus other object properties.

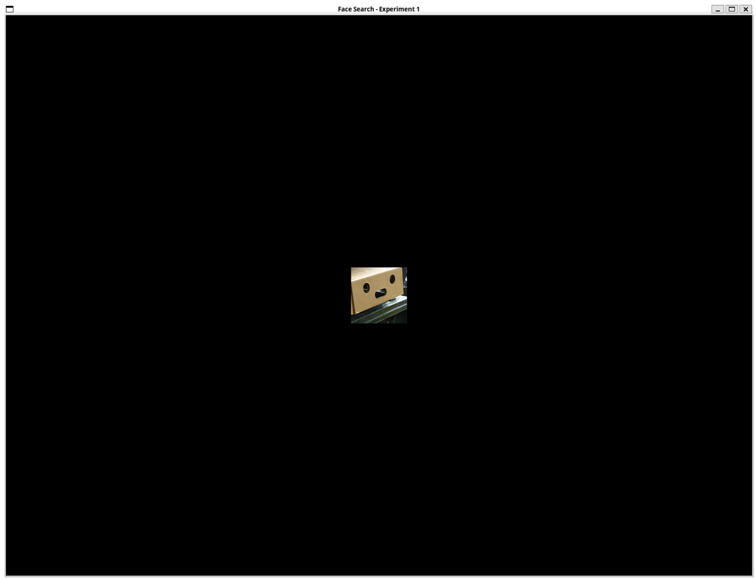


**Figure 1**

*Experiment 1 target image presentation (pFace variant).*


**Figure 2**

*Experiment 1 search array with target present in 8×8 grid layout.*


**Figure 3**

*Experiment 1 search array with target absent in 8×8 grid layout.*

## Experiment 2: Circular Layout (Heterogeneous Distractors)

### *Stimulus Presentation*

The trial sequence mirrors Experiment 1 with key spatial differences:

1. **Target Image**: Displayed for 1600 ms, with circular masking applied.
2. **Fixation**: 400–600 ms jittered fixation.
3. **Search Array**: Items arranged in an *equidistant circular layout* (radius ≈ 400 pixels, with all items ≈ 120 × 120 pixels). Set sizes are 4, 8, or 16.
4. **Feedback and Timeout**: Identical to Experiment 1.

### ***Experimental Factors***

| **Factor** | **Levels** | **N** |
|---|---|---|
| Target Type | 3 (nonFace, pFace, realFace) | |
| Target Presence | 2 | |
| Set Size | 3 (4, 8, 16) | |
| Object Categories | 23 | |
| Trial Types per Category | 18 | |
| **Total Trials** | | 414 |

Twelve participants completed Experiment 2. Participants were university students aged 20–22 years. Gender information was not recorded. All participants had normal or corrected-to-normal vision and provided informed consent. The dataset contains 828 trial rows, with 69 trials per participant.

### ***Heterogeneous Distractors***

Unlike Experiment 1, distractors in Experiment 2 come from **diverse object categories**, not the target's category. This increases task difficulty and allows differentiation between search slopes for different target types.

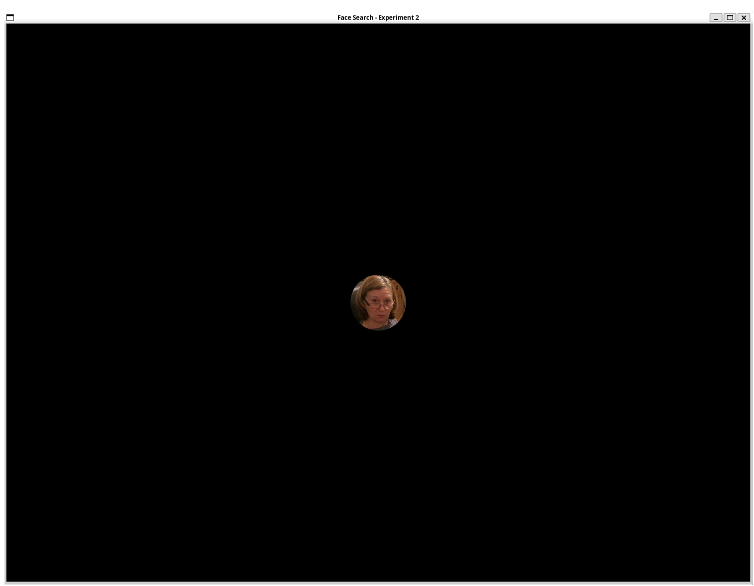


**Figure 4**
*Experiment 2 target image presentation (realFace variant with circular masking).*

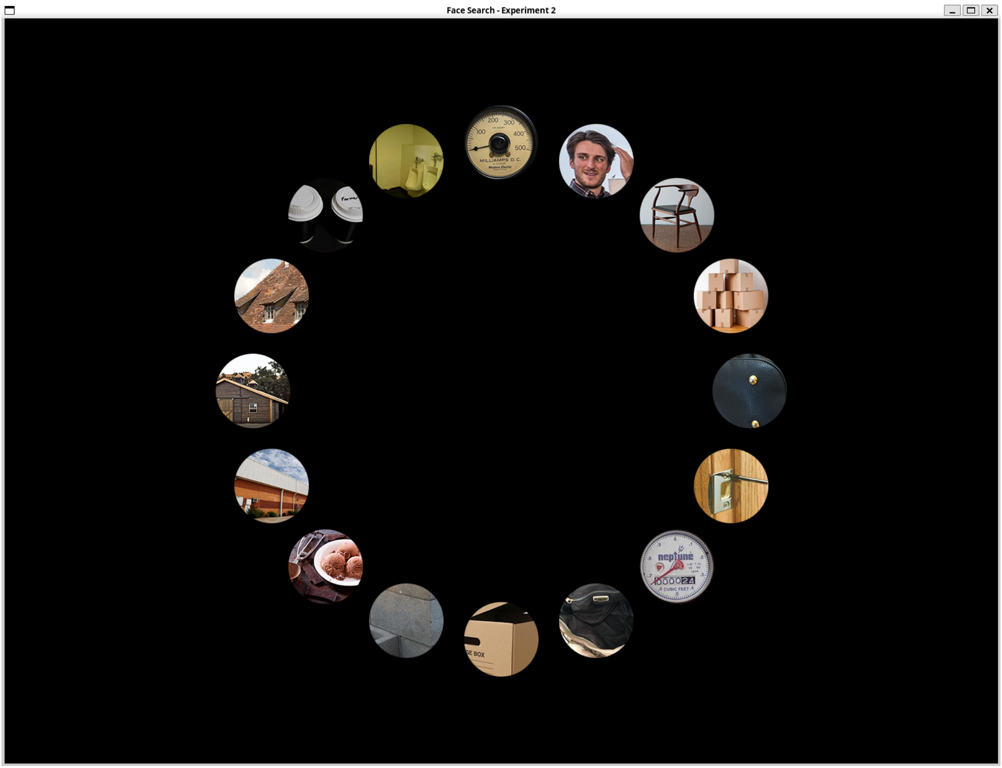


**Figure 5**

*Experiment 2 search array with heterogeneous distractors in circular layout.*

Due to the limited testing time available for each participant, we did not repeat the 12 (Experiment 1) or 18 (Experiment 2) trial types across multiple category blocks. Instead, Experiment 1 and Experiment 2 each used a single mixed trial set in which all condition combinations (set size × target presence × target type) were randomly interleaved. This adjustment changes only the repetition and blocking structure, while preserving the original factorial design and condition contrasts.

For data preprocessing, only correct-response trials were included in the analysis. Across all trials, 1,130 out of 1,352 trials were correct in Experiment 1, and 742 out of 828 trials were correct in Experiment 2.

## Data Analysis: Diffusion Decision Modeling

We analyzed trial-level reaction times (RTs) and binary present/absent responses using the diffusion decision model (DDM), implemented in Python with PyDDM. In DDM, decisions arise from noisy accumulation of evidence toward two response boundaries. Drift rate ($v$) reflects evidence accumulation efficiency, boundary separation ($a$) reflects response caution (speed–accuracy tradeoff), and nondecision time ($T_{er}$) captures perceptual and motor delays outside the decision process. Modeling full RT distributions provides stronger inference about underlying cognitive mechanisms than mean RTs alone.

Preprocessing followed standard psychophysical practice. Trials with missing RTs or timeouts were excluded, RTs were converted to seconds, and extreme RTs (below 0.20 s or above 10.00 s) were removed to reduce anticipatory responses and lapses. Reaction time preprocessing further excluded trials with response times exceeding 10,000 ms, consistent with standard psychophysical practice for removing lapses and delayed responses. After this additional exclusion, 1,072 trials remained in Experiment 1, while no additional trials were removed in Experiment 2 under this criterion. Models were fit to correct trials only, and responses were coded as present versus absent based on the trial condition and accuracy. These decisions focus the model on decision-related variability while minimizing contamination from nondecision artifacts.

Model fitting was conducted separately for each experiment, and within each experiment for target presence (present vs. absent) and target type. This separation reflects the expectation that target-absent trials can involve more exhaustive search and may rely on distinct decision processes. Two model structures were compared. Model A allowed drift rate to vary with set size while holding boundary separation constant, testing the hypothesis that larger arrays primarily reduce evidence efficiency. Model B allowed both drift rate and boundary separation to vary with set size, allowing response caution to increase with task difficulty. Drift rate was parameterized as a logarithmic function of set size, $v = v_0 + v_s \log(\text{set size})$, with a negative slope constraint to capture decreasing efficiency as arrays grow. Models were estimated by maximum-likelihood optimization and compared using log likelihood, Akaike Information Criterion (AIC; (Akaike, 1974)), and Bayesian Information Criterion (BIC; (Schwarz, 1978)), where lower values indicate better model fit after accounting for model complexity.

The logarithmic transformation of set size was chosen based on established properties of visual search behavior, where increases in display size produce diminishing marginal costs on performance (Wolfe, 1998). Empirically, reaction time functions in visual search are often better described by sublinear or logarithmic scaling rather than linear growth. From a modeling perspective, the log transform ensures that drift rate decreases monotonically with set size while avoiding unrealistic linear extrapolation at larger set sizes. This formulation is consistent with sequential sampling accounts of decision-making, in which increasing display complexity reduces evidence quality in a saturating rather than strictly proportional manner (Ratcliff & McKoon, 2008).

# Results

## Behavioral Results

Behavioral analyses revealed reliable effects of set size, target presence, and target category across both experiments. Reaction times increased systematically with larger set sizes, indicating increased search difficulty as visual competition increased. Target-absent trials were consistently slower than target-present trials, suggesting an additional cost associated with exhaustive search termination.

In Experiment 1, illusory-face targets were detected faster than matched nonface objects across conditions. In Experiment 2, reaction times followed a clear target-type hierarchy, with real faces producing the fastest responses, followed by illusory faces and then nonface objects. Accuracy remained relatively high across conditions, with only modest decreases at larger set sizes, indicating that the reaction-time effects were not driven by a speed–accuracy tradeoff.

Figure 6 shows the reaction time slopes by set size and target presence. The increase in reaction time with larger arrays is consistent with a serial visual search component, and the target-absent panels are uniformly higher, reflecting the cost of exhaustive search when the target is missing.

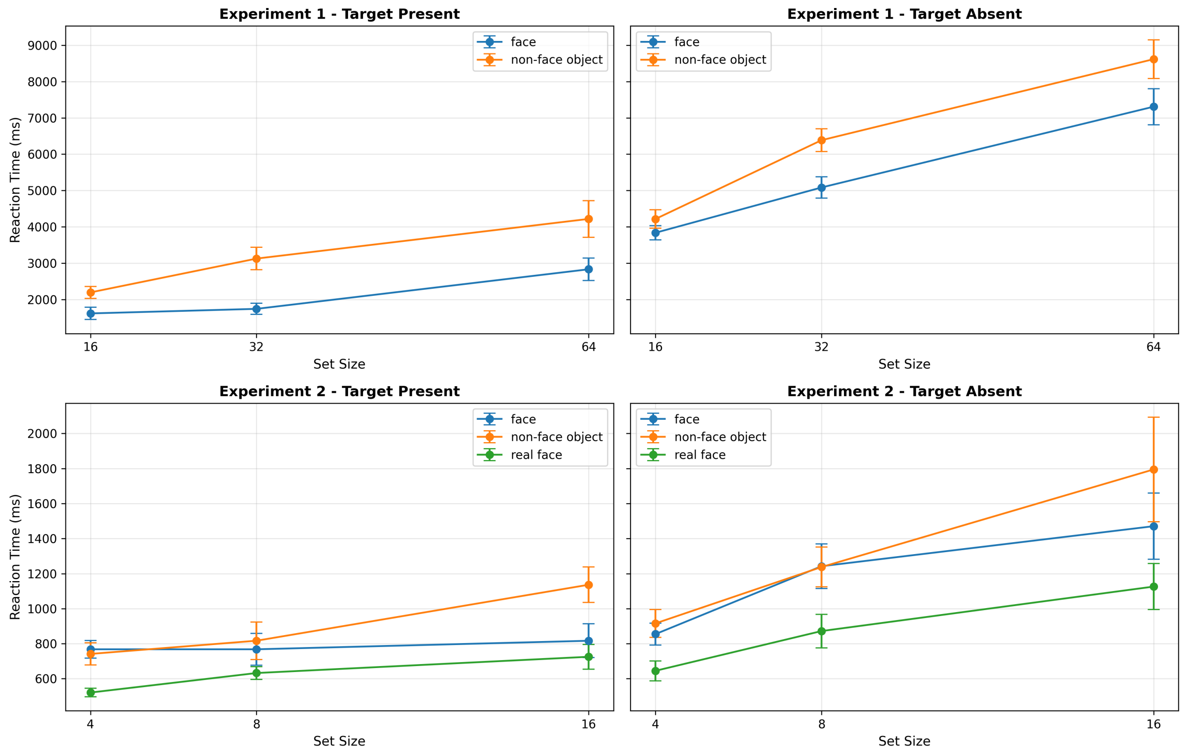


**Figure 6**

*Reaction time by set size and target presence.*

Figure 7 summarizes accuracy across the same conditions. Accuracy remains relatively high, with modest decreases at larger set sizes, indicating that the reaction time effects are not driven by a speed-accuracy tradeoff.

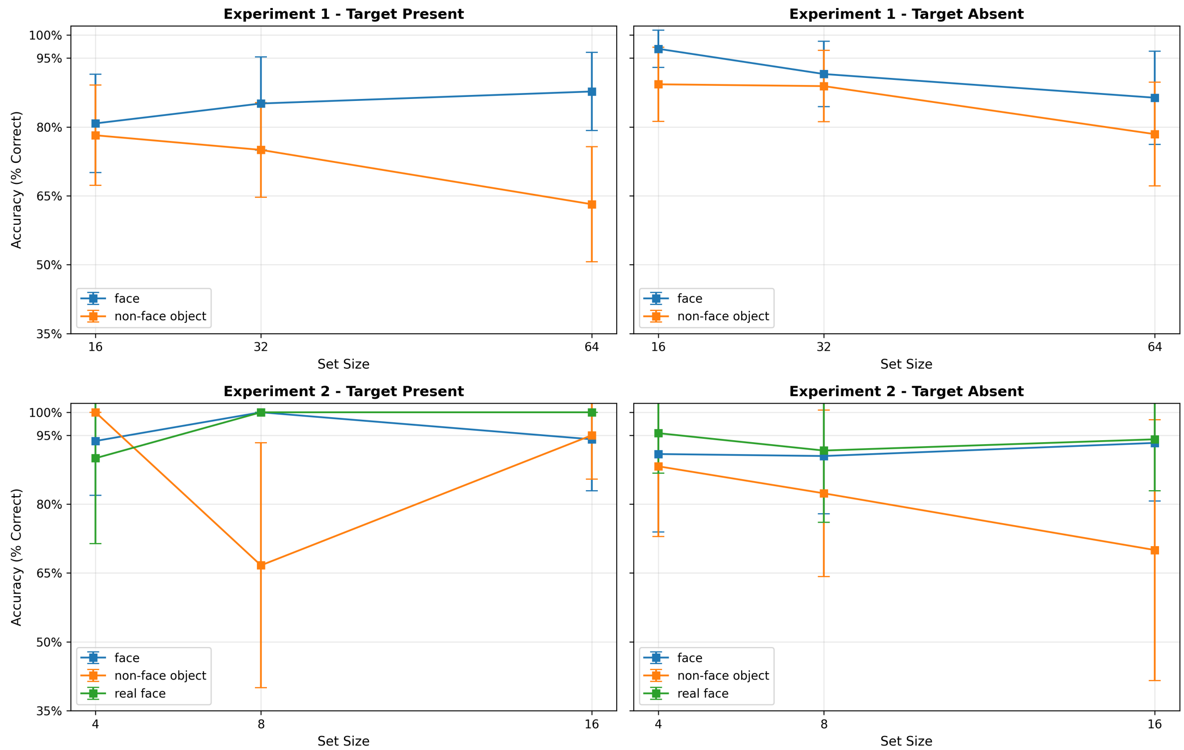


**Figure 7**

*Accuracy by set size and target presence.*

To isolate target presence effects, Figure 8 aggregates across set sizes. Target-absent trials show a consistent time penalty, and the ordering of target types is clear within each experiment, with illusory faces faster than non-face objects in Experiment 1 and real faces fastest in Experiment 2.

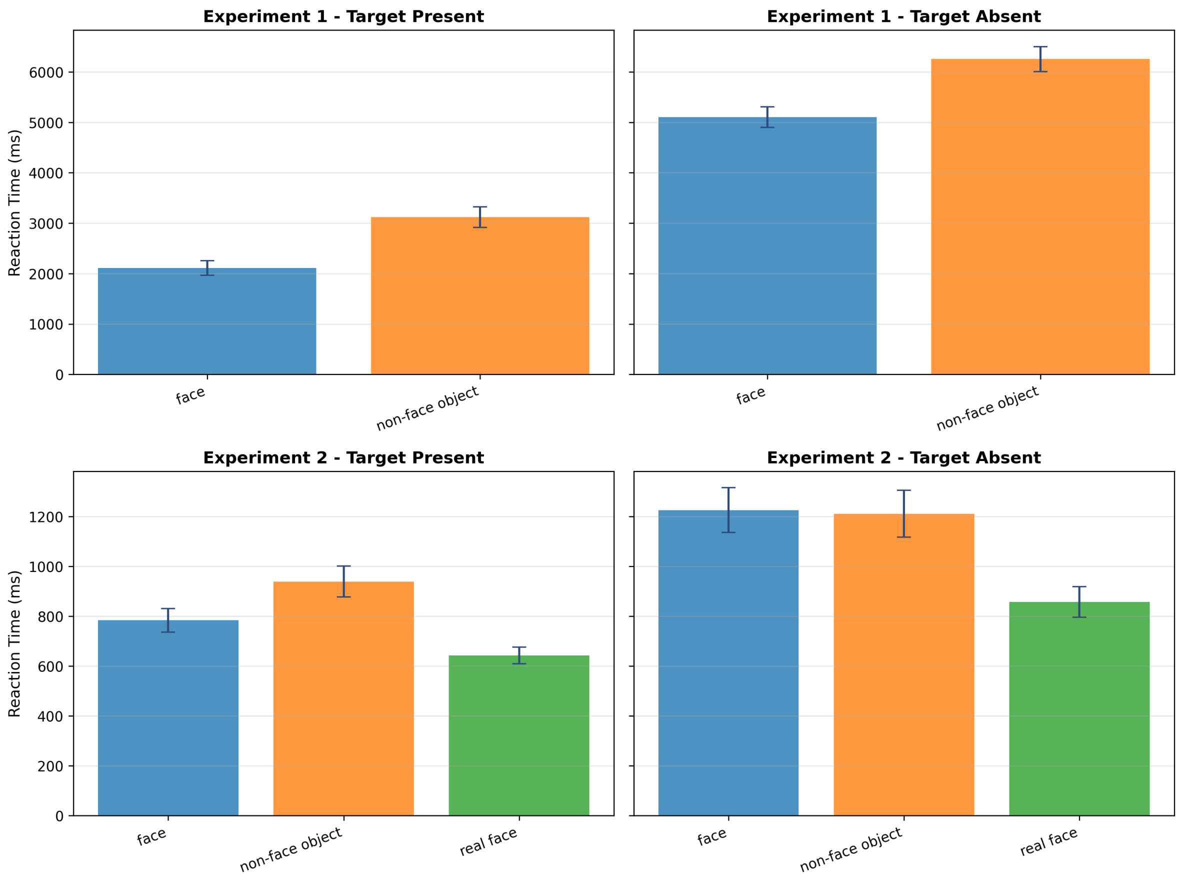


**Figure 8**

*Reaction time by target presence (aggregated across set sizes).*

Figure 9 shows accuracy aggregated across set sizes. The accuracy pattern mirrors the reaction time ordering, supporting the interpretation that faster target types are not simply due to lower decision thresholds.

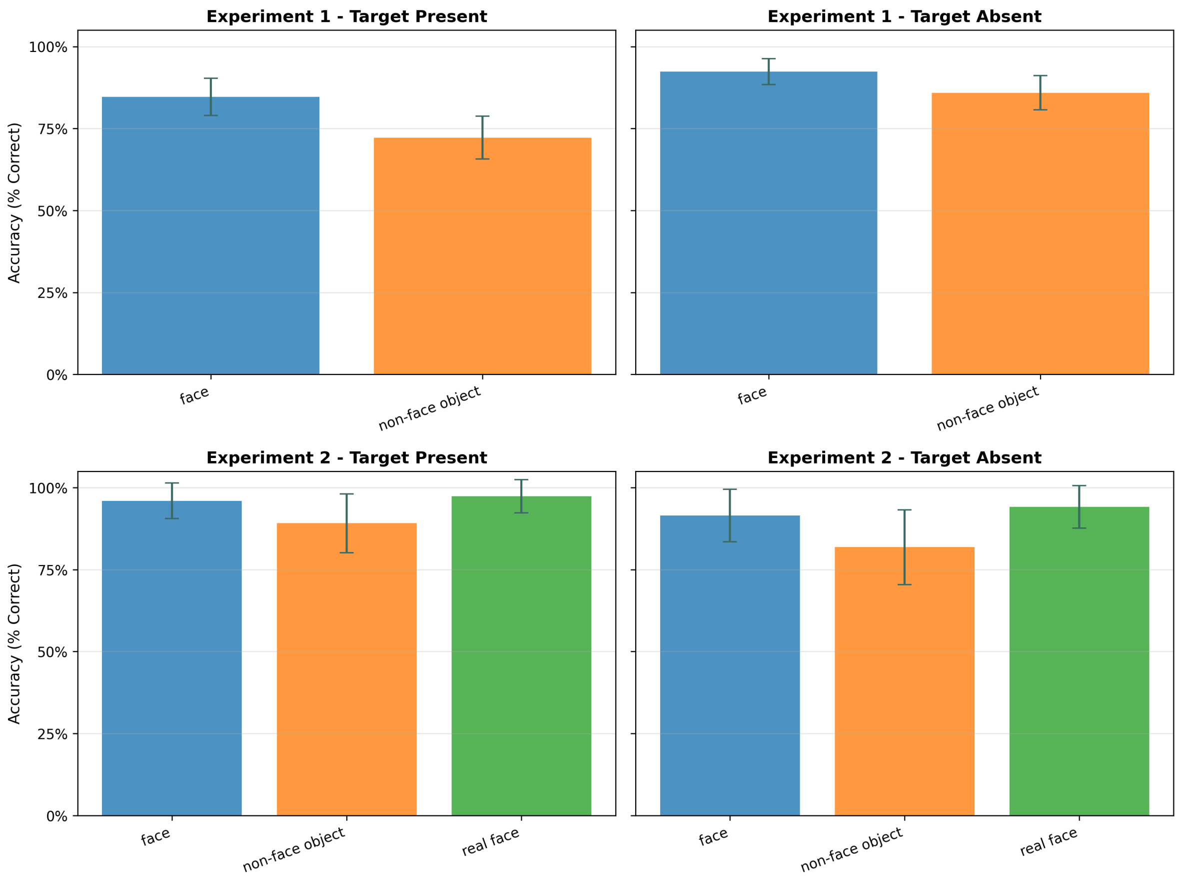


**Figure 9**

*Accuracy by target presence (aggregated across set sizes).*

Finally, Figure 10 contrasts overall reaction time and accuracy between experiments. Experiment 2 is faster overall, consistent with the smaller set sizes and heterogeneous distractors, while the target-type ordering remains stable across the aggregated comparison.

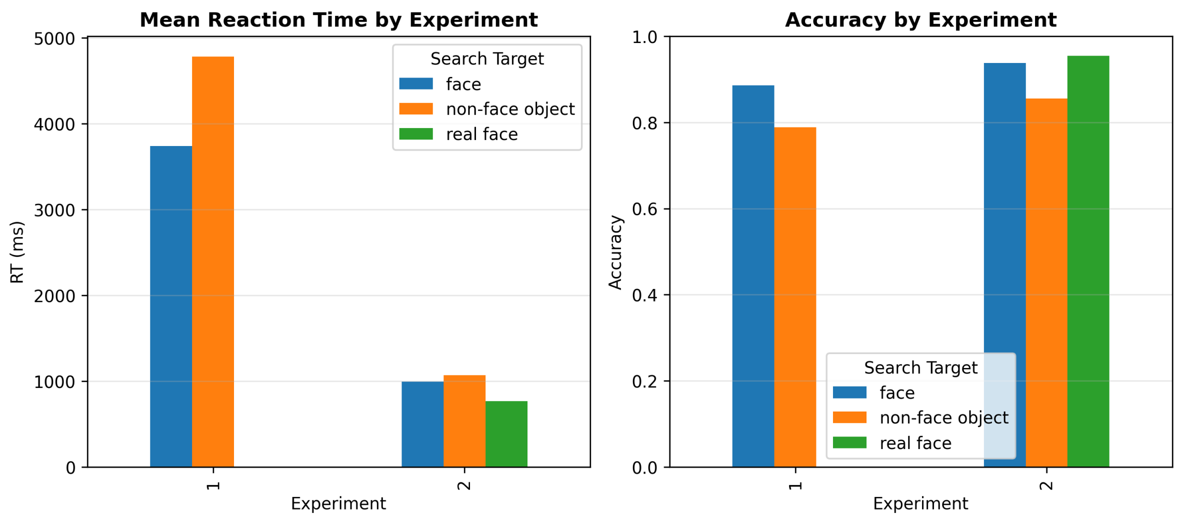


**Figure 10**

*Overall comparison of reaction time and accuracy by experiment.*

## Experiment 1: Diffusion Decision Modeling Results

Across target-present conditions, Model A consistently provided superior fit relative to Model B (Table 1), indicating that set-size effects were primarily driven by changes in drift rate rather than boundary separation. Drift rates decreased monotonically as set size increased for both illusory-face and nonface-object targets, suggesting that evidence accumulation became progressively less efficient under higher visual competition.

**Table 1**

*Model Comparison Results by Condition*

(a) *Target Present — Illusory Face ($n = 262$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 16 | 32 | 64 | 16 | 32 | 64 |
| A | 730.9324773 | 745.2058553 | 2.1079478 | 1.48769021 | 1.171704727 | 0.855719244 | | | |
| B | 732.1767936 | 753.5868606 | | 1.349100359 | 1.120738176 | 0.892375992 | 1.83969183 | 1.919181736 | 2.164598956 |

(b) *Target Present — Nonface Object ($n = 250$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 16 | 32 | 64 | 16 | 32 | 64 |
| A | 888.1125174 | 902.1983611 | 2.450479721 | 1.149582513 | 0.917655053 | 0.685727593 | | | |
| B | 892.0648127 | 913.1935782 | | 1.125841962 | 0.912405545 | 0.698969129 | 2.405688133 | 2.404304095 | 2.509238217 |

(c) *Target Absent — Illusory Face ($n = 286$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 16 | 32 | 64 | 16 | 32 | 64 |
| A | 1190.122353 | 1204.746321 | 3 | −0.760385571 | −0.760386264 | −0.760386957 | | | |
| B | 1183.294047 | 1205.229998 | | −0.732168914 | −0.732169607 | −0.732170301 | 2.509129713 | 3 | 3 |

(d) *Target Absent — Nonface Object ($n = 244$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 16 | 32 | 64 | 16 | 32 | 64 |
| A | 1148.613928 | 1162.602601 | 3 | −0.613250632 | −0.613251325 | −0.613252018 | | | |
| B | 1149.023122 | 1170.006131 | | −0.596934775 | −0.596935468 | −0.596936161 | 2.715069377 | 3 | 3 |

*Note.* Lower AIC and BIC values indicate better model fit.
$v$ = drift rate; $a$ = boundary separation.
$v_{16}$, $v_{32}$, and $v_{64}$ represent estimated drift rates for set sizes 16, 32, and 64, respectively.
Fixed Boundary ($a$) indicates a single boundary parameter shared across set sizes in Model A, whereas $a_{16}$, $a_{32}$, and $a_{64}$ represent boundary estimates that varied with set size in Model B.

Importantly, illusory-face targets produced consistently higher drift rates than nonface-object targets across all set sizes. This pattern suggests that illusory faces benefited from enhanced evidence accumulation relative to matched nonface objects, consistent with partial recruitment of face-related perceptual processing during visual search.

Boundary estimates in Model B showed comparatively weak or inconsistent modulation across set sizes in target-present conditions. Moreover, Model B did not improve model fit according to either AIC or BIC, further supporting a drift-dominated account of target-present decisions.

Observed-versus-simulated RT distributions for the target-present conditions are shown in Figures 11–12.

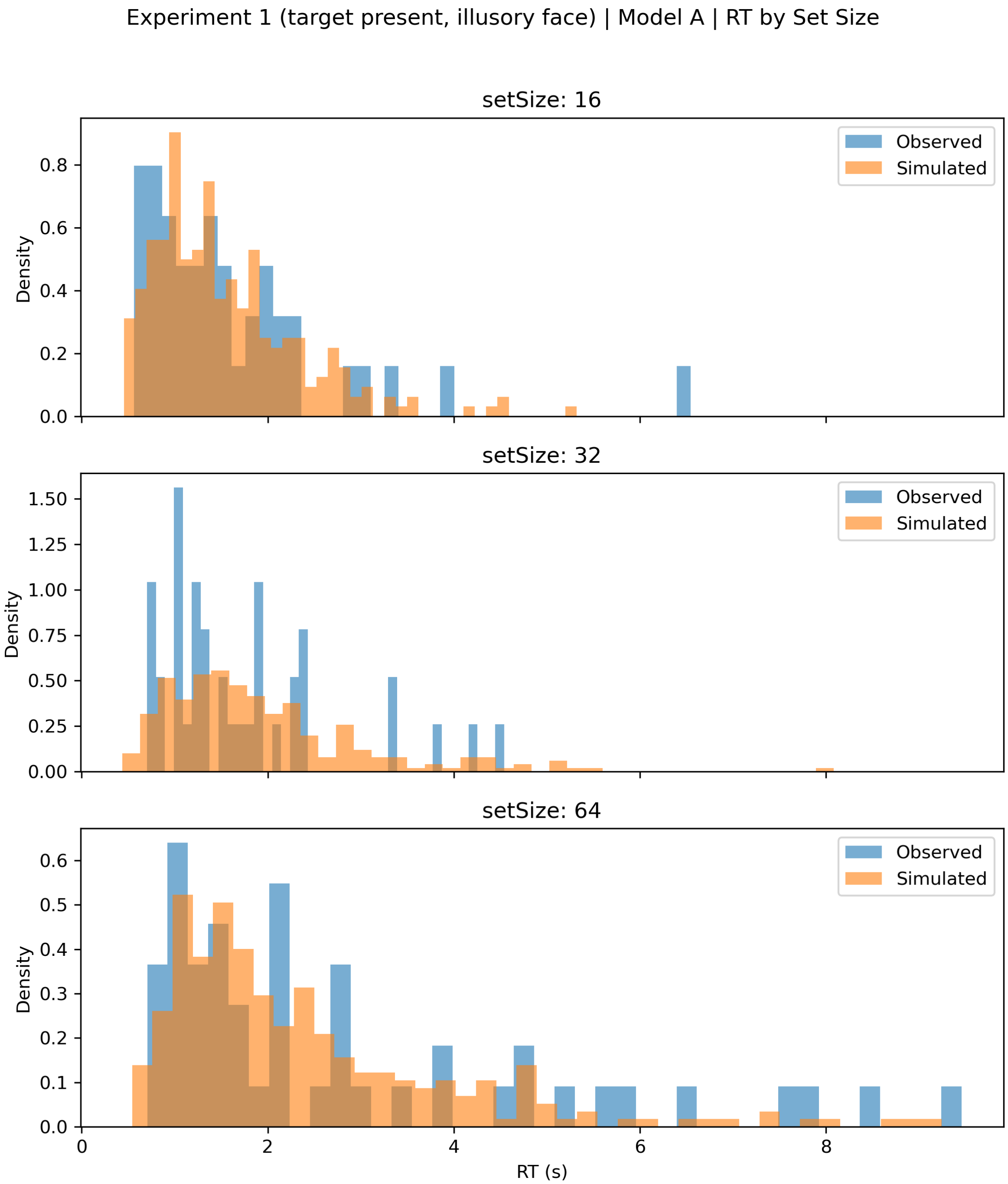


**Figure 11**

*Observed and simulated RT distributions across set sizes for the target-present illusory-face condition under Model A.*

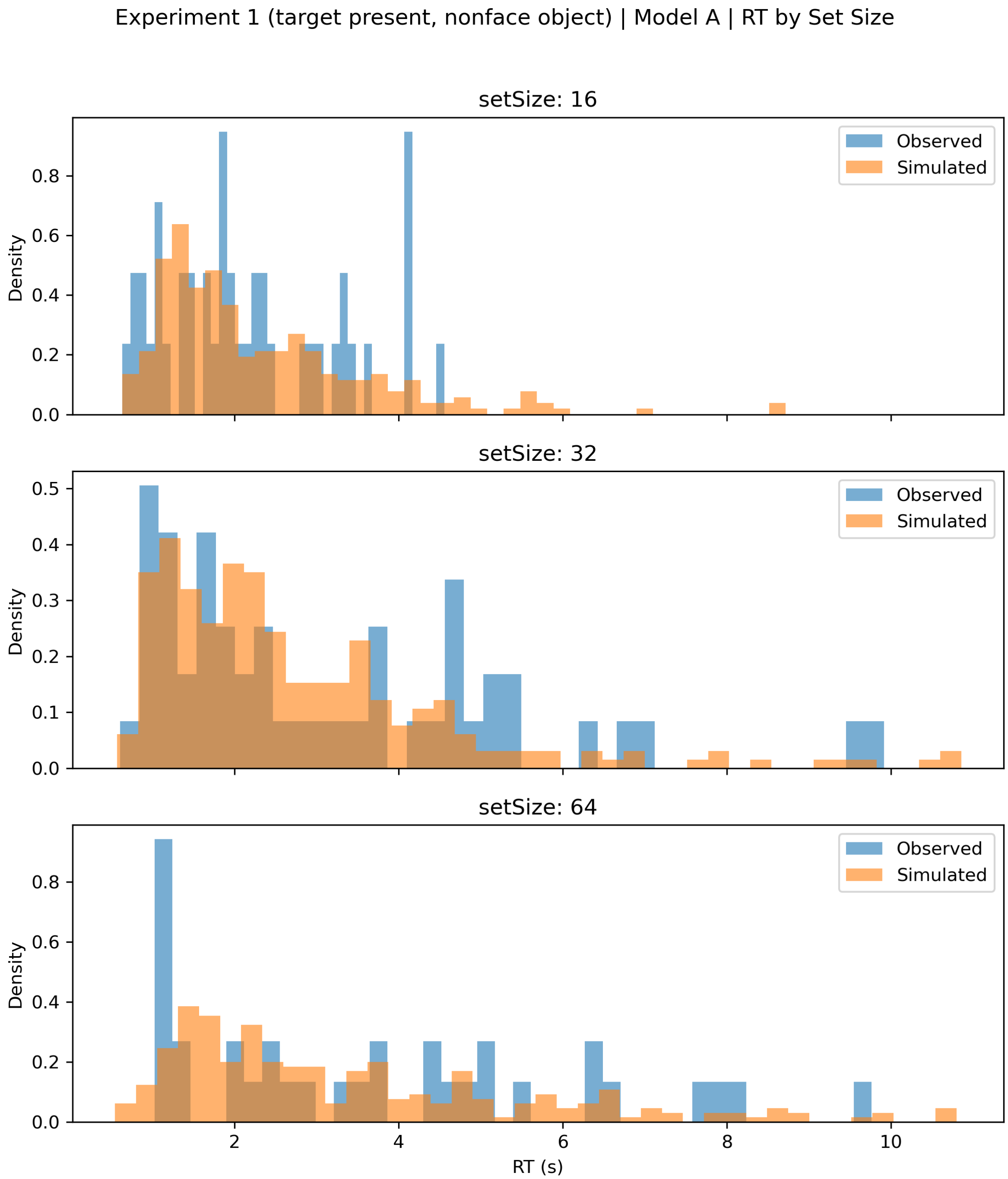


**Figure 12**

*Observed and simulated RT distributions across set sizes for the target-present nonface-object condition under Model A.*

In contrast, target-absent conditions showed a different computational pattern. Drift estimates remained largely stable across set sizes, whereas boundary separation increased substantially in Model B. In both target categories, Model B yielded lower AIC values than Model A, suggesting that target-absent decisions were better characterized by increased response caution than by changes in evidence accumulation efficiency.

Boundary estimates approached the upper fitting limit at larger set sizes in the target-absent conditions, consistent with increasingly conservative search termination processes when no target evidence was detected.

Observed-versus-simulated RT distributions for the target-absent conditions are shown in Figures 13–14.

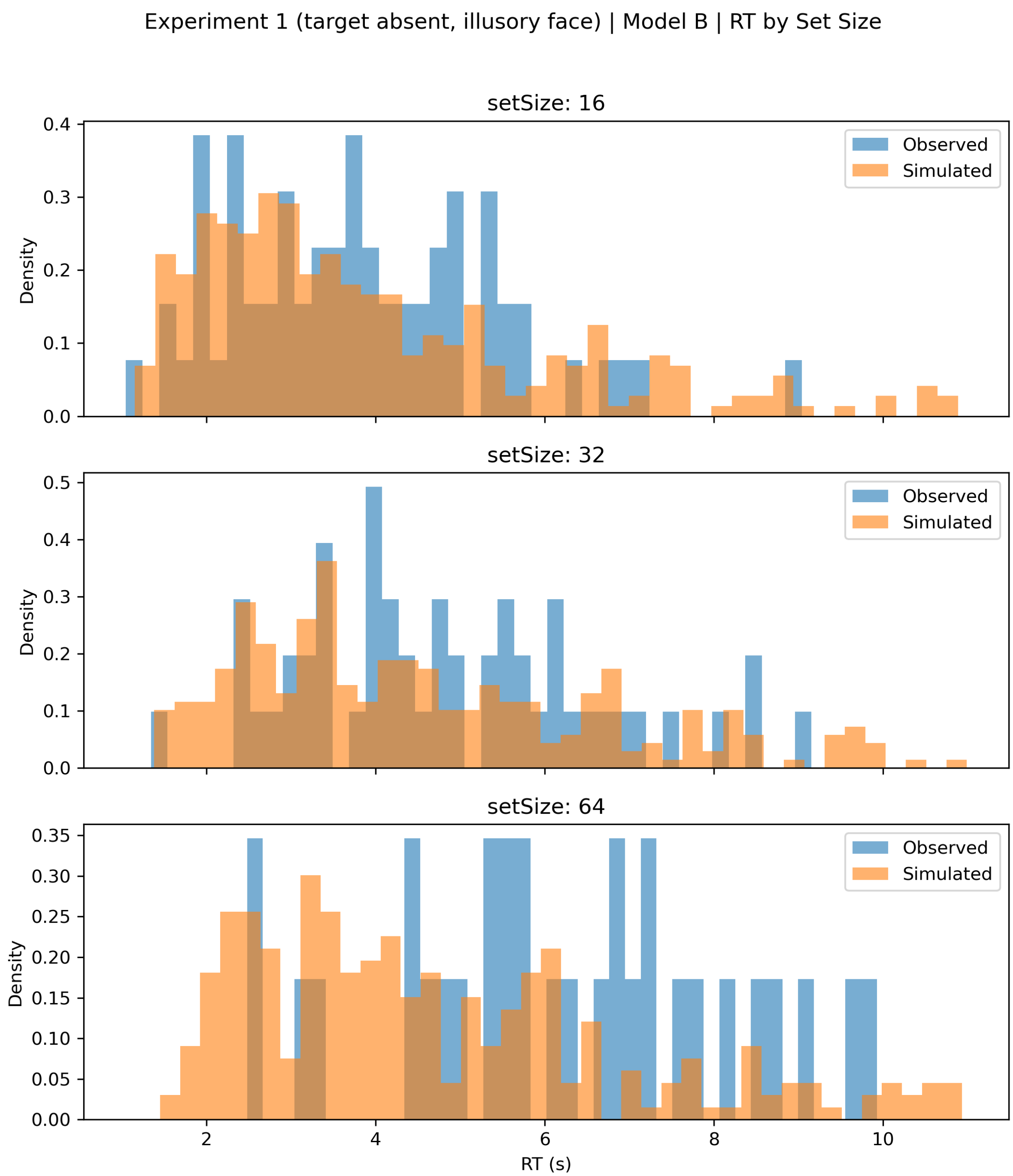


**Figure 13**

*Observed and simulated RT distributions across set sizes for the target-absent illusory-face condition under Model B.*

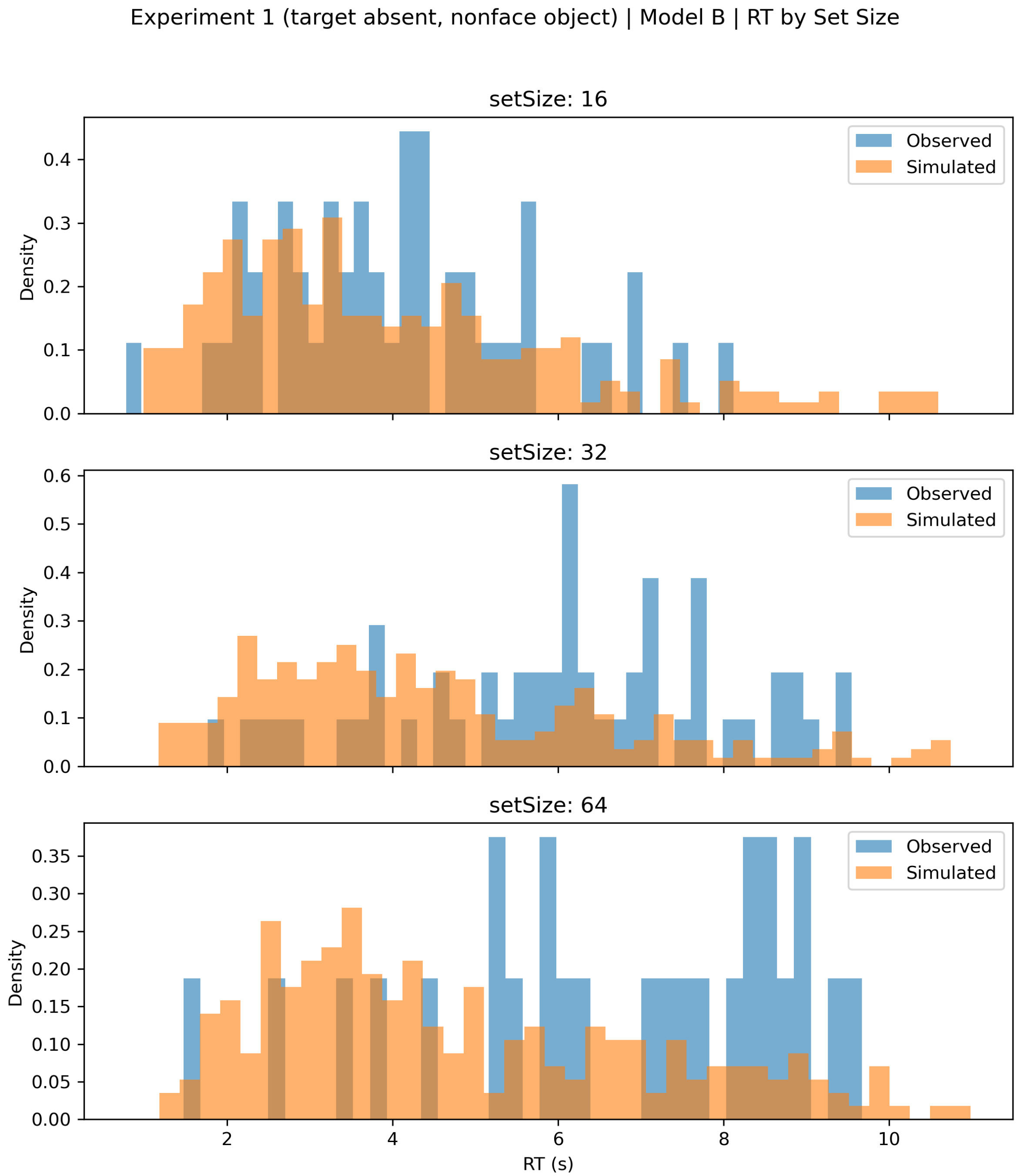


**Figure 14**

*Observed and simulated RT distributions across set sizes for the target-absent nonface-object condition under Model B.*

**Experiment 2: Diffusion Decision Modeling Results**

Across target-present conditions, model comparison consistently favored Model A over Model B (Table 2), indicating that set-size effects were primarily drift-driven rather than boundary-driven. Drift rates decreased monotonically with set size for all three target categories, demonstrating reduced evidence accumulation efficiency as visual competition increased.

**Table 2**

*Model Comparison Results by Condition (Set 2)*

(a) *Target Present — Illusory Face ($n = 130$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 4 | 8 | 16 | 4 | 8 | 16 |
| A | 48.62839018 | 60.09852798 | 1.803553248 | 2.793189578 | 2.575624927 | 2.358060277 | | | |
| B | 50.22100134 | 67.42620804 | | 2.892086336 | 2.607235933 | 2.322385531 | 1.942302779 | 1.700413432 | 1.840539119 |

(b) *Target Present — Nonface Object ($n = 119$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 4 | 8 | 16 | 4 | 8 | 16 |
| A | 41.50293775 | 52.61943172 | 2.108337841 | 3.450788945 | 2.779340413 | 2.107891881 | | | |
| B | 45.30006326 | 61.97480422 | | 3.400118355 | 2.777470247 | 2.154822139 | 2.089532818 | 2.062038859 | 2.171331687 |

(c) *Target Present — Real Face ($n = 127$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 4 | 8 | 16 | 4 | 8 | 16 |
| A | −99.11420672 | −87.73745837 | 1.012435469 | 3.426455892 | 3.011325965 | 2.596196039 | | | |
| B | −97.56207146 | −80.49694894 | | 3.111453981 | 2.991914583 | 2.872375185 | 0.892056771 | 1.008847184 | 1.104410646 |

(d) *Target Absent — Illusory Face ($n = 119$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 4 | 8 | 16 | 4 | 8 | 16 |
| A | 170.7697812 | 181.8862752 | 2.073019173 | −1.893574628 | −1.893575321 | −1.893576014 | | | |
| B | 165.0858815 | 181.7606225 | | −1.984095415 | −1.984096109 | −1.984096802 | 1.902936725 | 2.0618256 | 2.510511442 |

(e) *Target Absent — Nonface Object ($n = 115$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 4 | 8 | 16 | 4 | 8 | 16 |
| A | 168.3542647 | 179.3339932 | 2.145092637 | −1.867076819 | −1.867077513 | −1.867078206 | | | |
| B | 130.9772599 | 147.4468527 | | −2.054250928 | −2.054251621 | −2.054252314 | 1.540439938 | 2.106553621 | 3 |

(f) *Target Absent — Real Face ($n = 132$)*

| Model | AIC | BIC | Fixed Boundary (a) | Drift Rate (v) | | | Boundary Separation (a) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 4 | 8 | 16 | 4 | 8 | 16 |
| A | 72.75185046 | 84.28305815 | 1.655677944 | −2.304984501 | −2.304985194 | −2.304985887 | | | |
| B | 23.9942143 | 41.29102583 | | −2.573665499 | −2.573666192 | −2.573666885 | 1.096156743 | 1.716749431 | 2.118693971 |

*Note.* Lower AIC and BIC values indicate better model fit.
$v$ = drift rate; $a$ = boundary separation.
$v_4$, $v_8$, and $v_{16}$ represent estimated drift rates for set sizes 4, 8, and 16, respectively.
Fixed Boundary ($a$) indicates a single boundary parameter shared across set sizes in Model A, whereas $a_4$, $a_8$, and $a_{16}$ represent boundary estimates that varied with set size in Model B.

Importantly, drift estimates revealed a clear target-type hierarchy. Real faces produced the highest drift rates across all set sizes, followed by illusory faces and then nonface objects. Real-face drift rates remained comparatively high even at the largest set size, suggesting the strongest robustness to increasing search difficulty. Illusory faces showed intermediate drift rates, indicating partial recruitment of face-related perceptual processing, whereas nonface objects exhibited the lowest drift rates and the steepest decline as set size increased.

Boundary estimates in Model B showed no consistent monotonic pattern across target-present conditions, and Model B did not improve model fit relative to Model A. Together, these findings support a drift-dominated account of target-present search decisions in Experiment 2.

Observed-versus-simulated RT distributions for the three target-present conditions are shown in Figures 15–17.

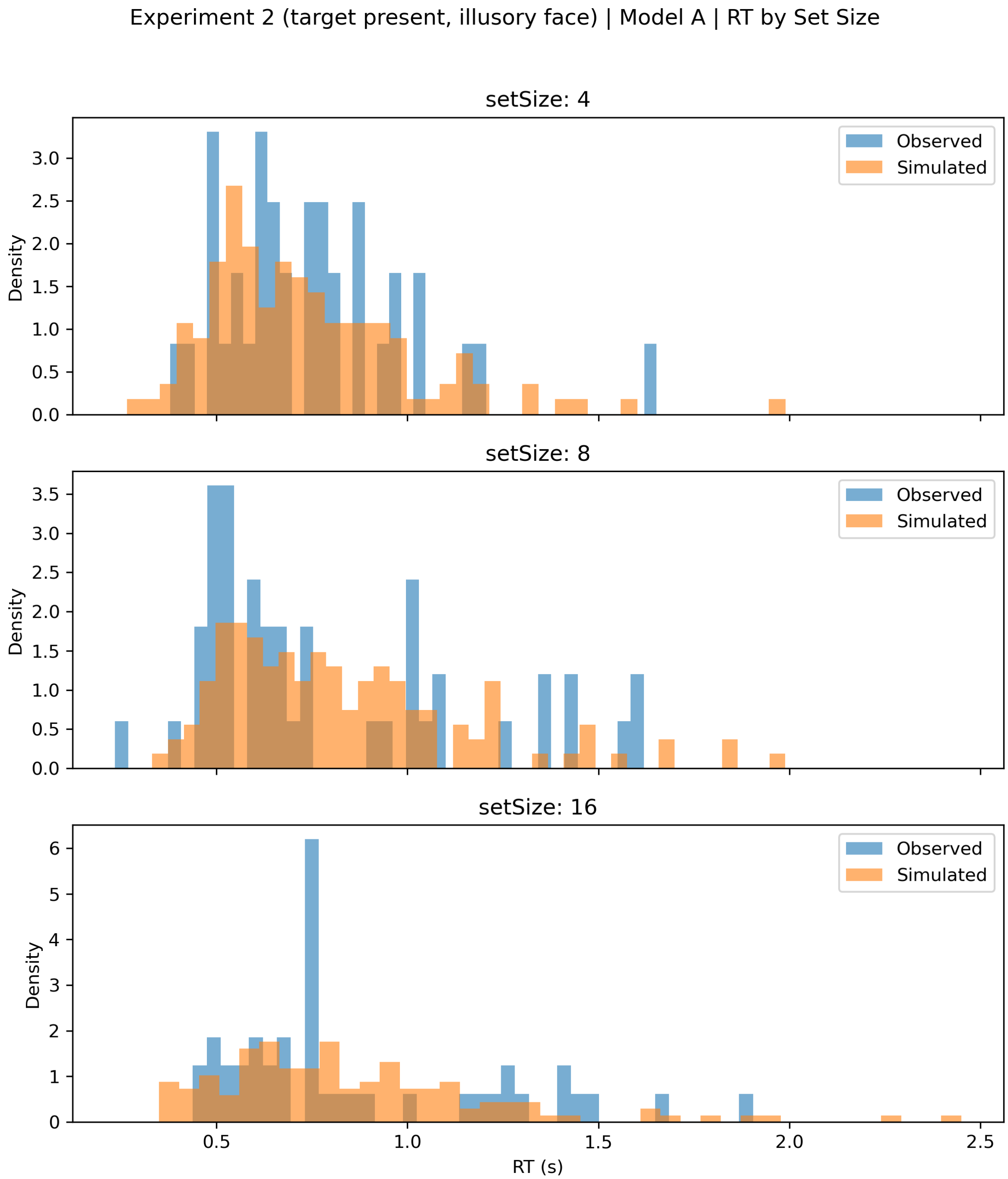


**Figure 15**

*Observed and simulated RT distributions across set sizes for the target-present illusory-face condition under Model A.*

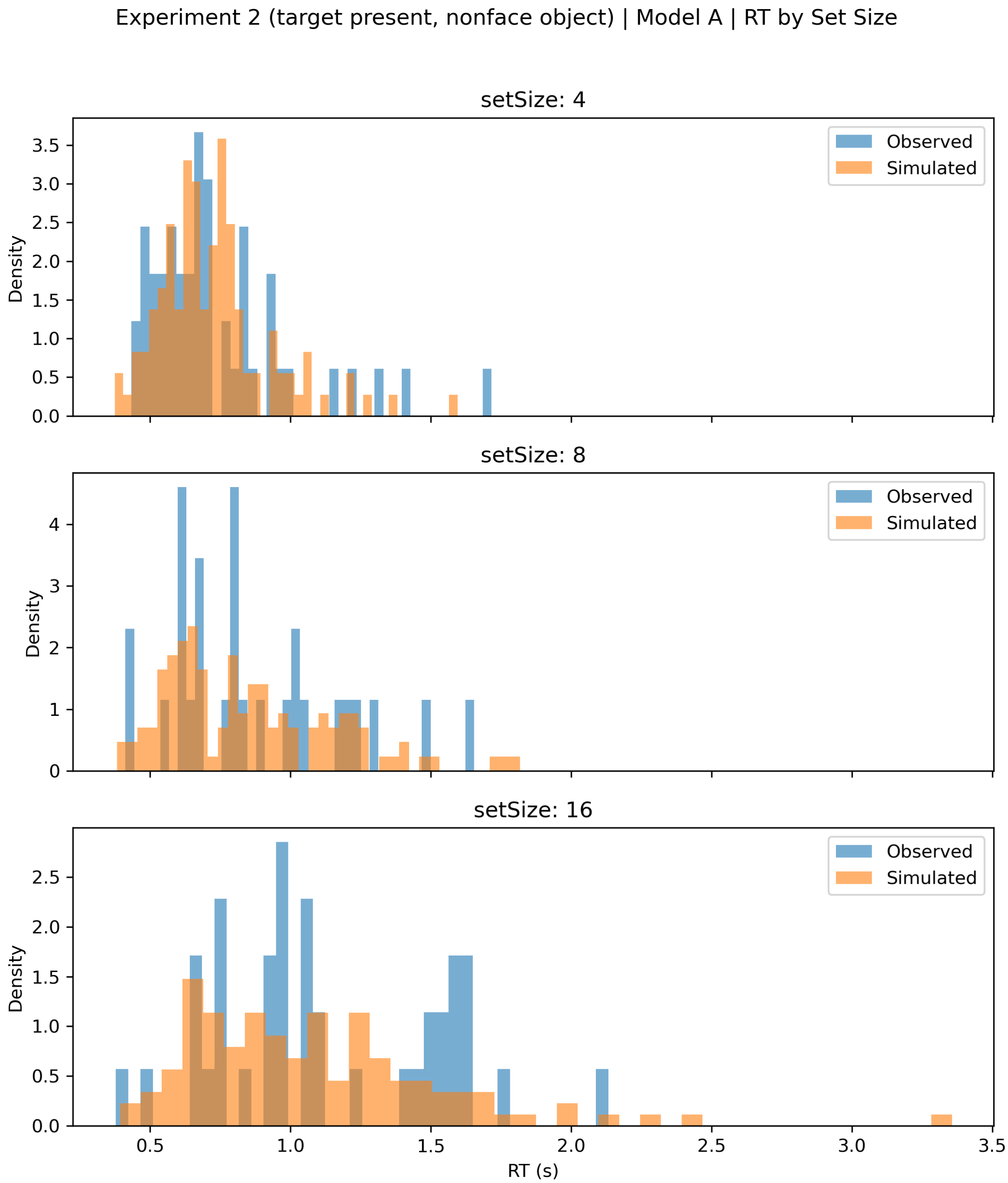


**Figure 16**

*Observed and simulated RT distributions across set sizes for the target-present nonface-object condition under Model A.*

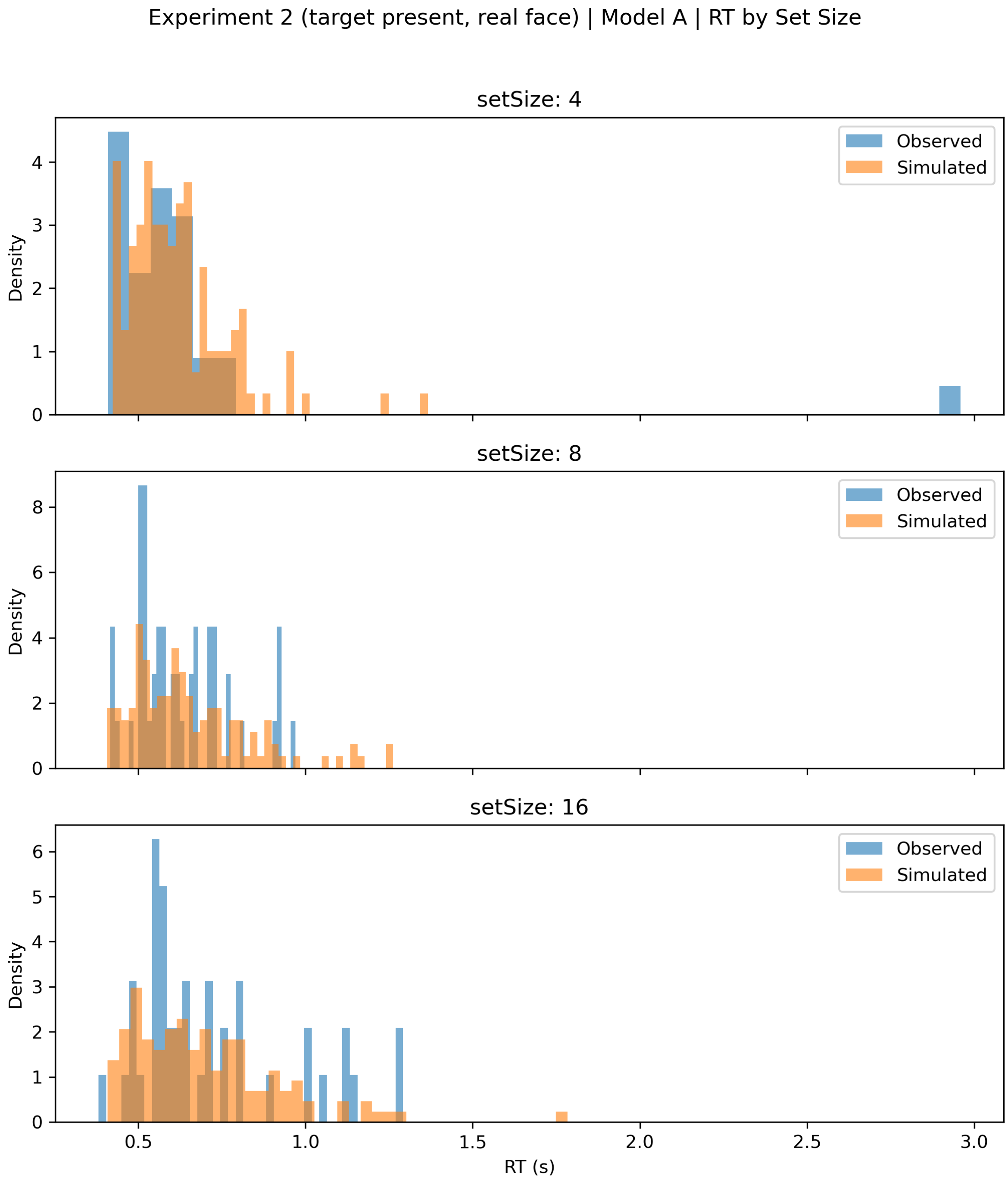


**Figure 17**

*Observed and simulated RT distributions across set sizes for the target-present real-face condition under Model A.*

In contrast, target-absent conditions showed a different pattern. Across all three target categories, Model B generally provided superior fit metrics compared with Model A

(Table 2). Drift estimates remained relatively stable across set sizes, whereas boundary separation increased systematically as array size increased. This pattern suggests that target-absent judgments were governed primarily by increasingly conservative response criteria rather than changes in evidence accumulation efficiency.

The strongest boundary modulation was observed for real-face and nonface-object target-absent conditions, in which boundary estimates approached the upper fitting limit at larger set sizes. This finding is consistent with a cautious search termination process in the absence of detectable target evidence.

Observed-versus-simulated RT distributions for the target-absent conditions under Model B are shown in Figures 18–20.

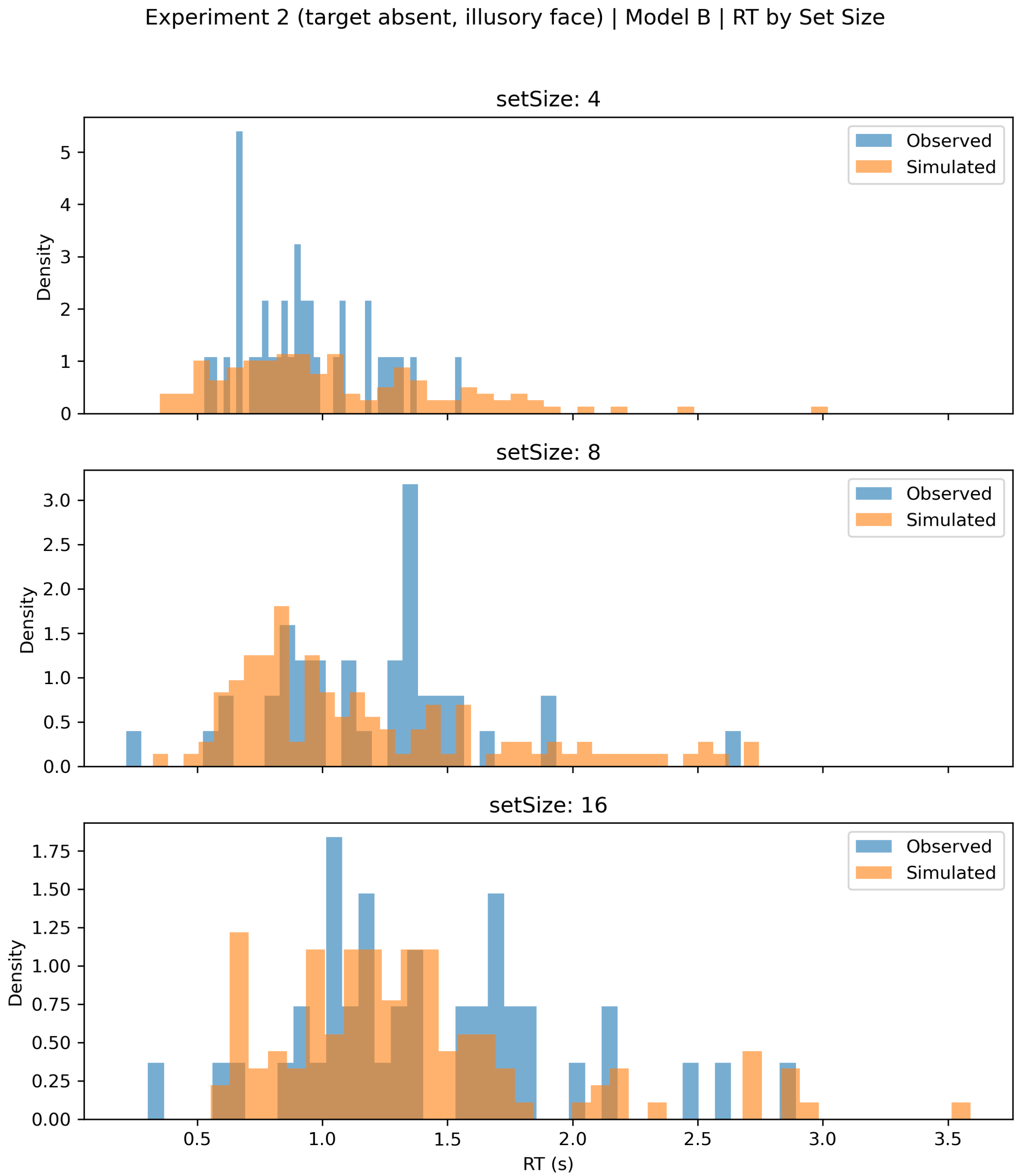


**Figure 18**

*Observed and simulated RT distributions across set sizes for the target-absent illusory-face condition under Model B.*

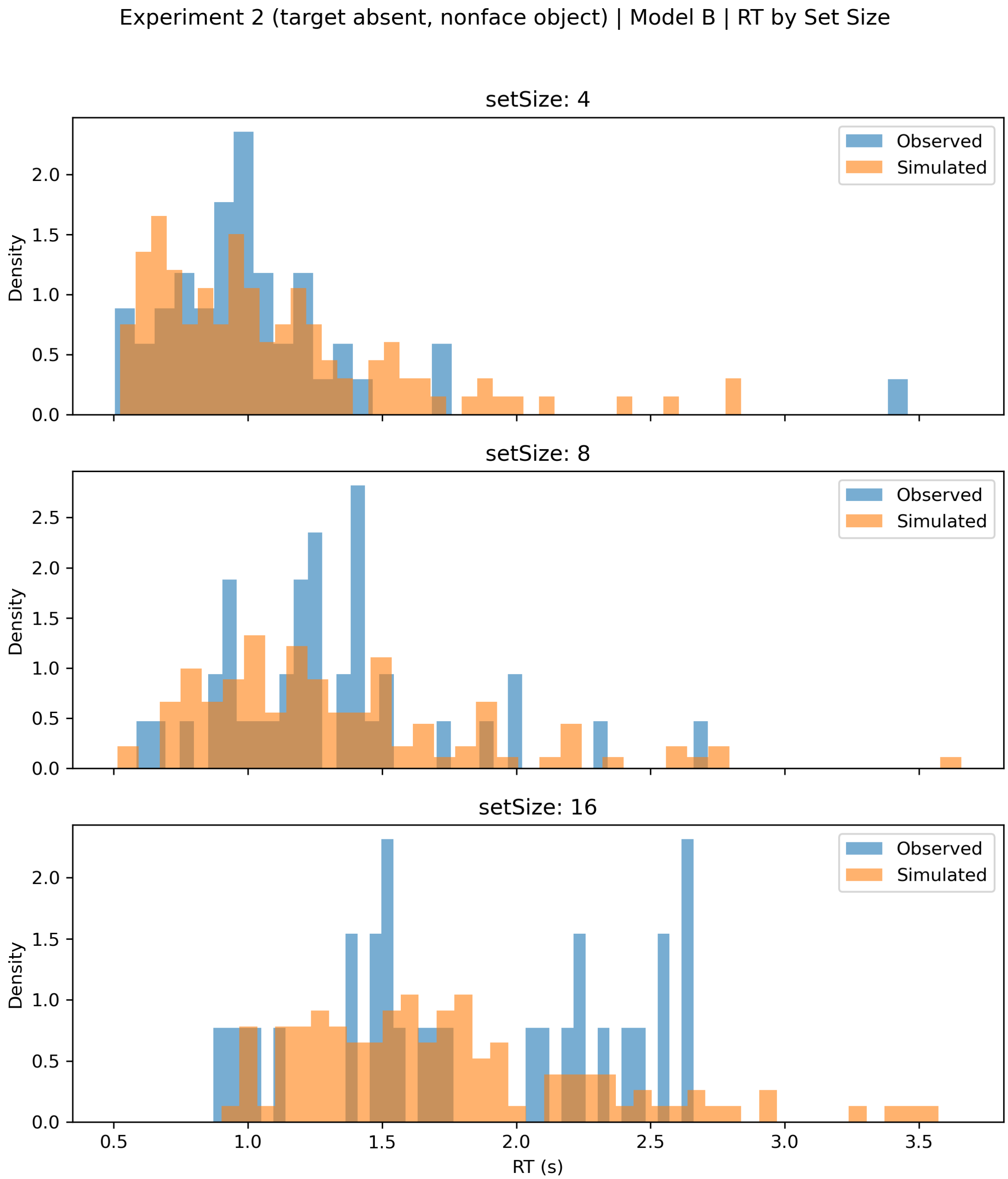


**Figure 19**

*Observed and simulated RT distributions across set sizes for the target-absent nonface-object condition under Model B.*

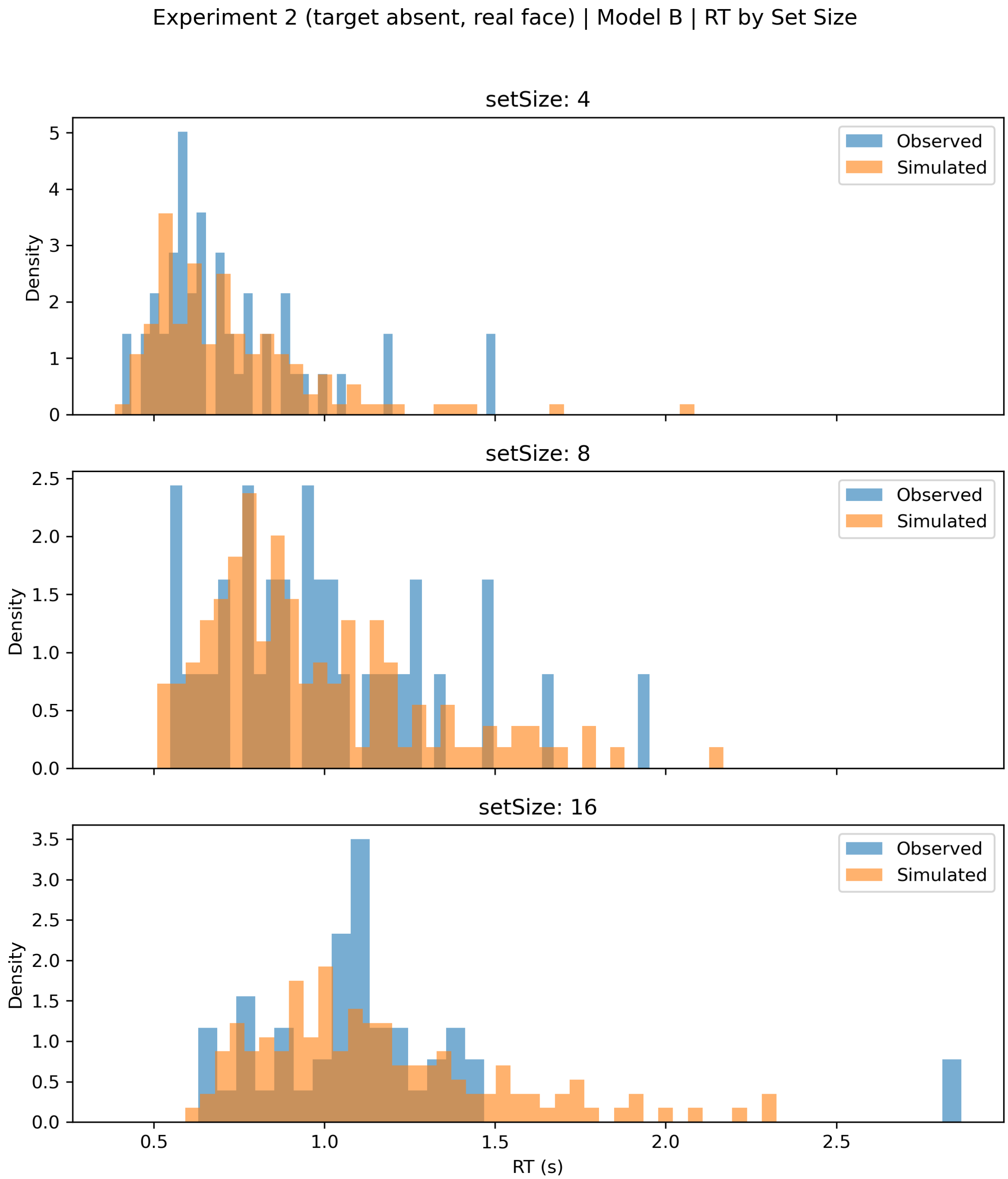


**Figure 20**

*Observed and simulated RT distributions across set sizes for the target-absent real-face condition under Model B.*

## Discussion

The present findings broadly supported the hypotheses proposed in the Introduction. Across both experiments, illusory-face stimuli produced faster responses and higher drift rates than matched nonface objects, while real faces produced the strongest overall search advantage. These results suggest that the visual-search advantage for illusory faces primarily reflects enhanced evidence accumulation rather than reduced response caution.

This pattern is broadly consistent with previous diffusion-model studies of face perception. For example, Tu et al. (2020) reported that perceptual decisions toward faces are associated with enhanced evidence accumulation, reflected by increased drift rates during face categorization tasks. The present findings extend this account by showing that similar drift-rate advantages emerge not only for real human faces, but also for illusory face configurations embedded in everyday objects.

Importantly, the intermediate drift rates observed for illusory faces suggest that face-related perceptual mechanisms can be partially recruited even in the absence of genuine facial structure. This supports the idea that the visual system is highly sensitive to coarse facial configuration and may prioritize face-like patterns during perceptual decision-making.

By contrast, target-absent decisions were primarily characterized by increased boundary separation, suggesting more conservative search termination processes as display complexity increased. This dissociation between drift-rate modulation and boundary adjustment highlights the utility of DDM for separating perceptual evidence accumulation from strategic response caution in visual search tasks.

## Conclusion

This study reproduced the illusory-face visual-search advantage reported by Keys et al. (2021) and extended the paradigm using diffusion decision modeling. The findings suggest that illusory faces benefit from enhanced evidence accumulation during visual search, positioning pareidolia as a meaningful component of face-related perceptual processing. More broadly, the study demonstrates how DDM can reveal latent cognitive mechanisms underlying reaction-time effects in complex visual search behavior.

## References

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, *19*(6), 716–723. https://doi.org/10.1109/TAC.1974.1100705

Hershler, O., & Hochstein, S. (2005). At first sight: A high-level pop out effect for faces. *Vision Research*, *45*(13), 1707–1724. https://doi.org/10.1016/j.visres.2005.01.015

Keys, R. T., Taubert, J., & Wardle, S. G. (2021). A visual search advantage for illusory faces in objects. *Attention, Perception, & Psychophysics*, *83*, 1942–1953. https://doi.org/10.3758/s13414-021-02267-4

Ratcliff, R., & McKoon, G. (2008). The diffusion decision model: Theory and data for two-choice decision tasks. *Neural Computation*, *20*(4), 873–922. https://doi.org/10.1162/neco.2008.12-06-420

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, *6*(2), 461–464. https://doi.org/10.1214/aos/1176344136

Tu, T., Wang, Y., Philiastides, M. G., & Sajda, P. (2020). Simultaneous EEG-fMRI study reveals separate bias mechanisms modulated by stimulus evidence during rapid face processing. *The Journal of Neuroscience*, *40*(6), 1244–1252. https://doi.org/10.1523/JNEUROSCI.1187-19.2019

Wolfe, J. M. (2001). Asymmetries in visual search: An introduction. *Perception & Psychophysics*, *63*(3), 381–389. https://doi.org/10.3758/BF03194406

Wolfe, J. M. (1998). Visual search. In H. Pashler (Ed.), *Attention* (pp. 13–73). Psychology Press.

# Author Notes

## Project Duration

The project was completed over an estimated total period of approximately 4–6 weeks. The main components of the workflow included implementation, data collection, preprocessing, modeling, and report writing.

The initial translation of the experimental paradigm from MATLAB to Python required approximately 15 hours of development and debugging. Data collection was conducted over approximately 2 days of experimental sessions. Data preprocessing and visualization required approximately 5 hours, while diffusion decision modeling and parameter estimation required approximately 30 hours. The final report writing and revision process required approximately 40 hours.

## AI Usage Statement

AI-assisted tools were extensively used throughout this project, particularly for software development. A substantial portion of the initial Python/Pygame implementation was generated with AI assistance, including stimulus presentation logic, experimental control flow, and data handling scripts. These outputs were subsequently reviewed, debugged, and modified by the authors to ensure correctness and alignment with the experimental design.

In addition, AI tools were used to support understanding of diffusion decision modeling (DDM), assist with LaTeX formatting, and provide suggestions for language refinement.

All final experimental design decisions, data preprocessing steps, and modeling choices were independently verified and validated by the authors.